Hormones in perinatal rat and spiny mouse: relation to altricial and precocial timing of birth

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LAMERS, WOUTER H., PIET G. MOOREN, HANS GRIEP, ERIK ENDERT, HERMAN J. DEGENHART, AND ROBERT CHARLES. *Hormones in perinatal rat and spiny mouse: relation to altricial and precocial timing of birth.* Am. J. Physiol. 251 (Endocrinol. Metab. 14): E78–E85, 1986.—Rat (*Rattus norvegicus*) and spiny mouse (*Acomys cahirinus*) are closely related murine species that, due to their altricial (rat) and precocial (spiny mouse) modes of development, differ in the developmental timing of birth. A comparison between the developmental profiles of plasma glucagon, insulin, thyroxine, triiodothyronine, and glucocorticosteroid hormone was carried out to elucidate the question to what extent these hormonal profiles were related to the timing of birth. Although corticosterone is the major circulating glucocorticosteroid in rat, only cortisol was found in the spiny mouse. The onset of increases in glucocorticosteroid and thyroid hormone levels occurred at the same developmental time points in both species. A neonatal increase in triiodothyronine levels was observed in the spiny mouse only. In both species the immediate perinatal period was characterized by decreases in the ratio of insulin and glucagon levels and the level of glucocorticosteroids. The observed developmental patterns of hormonal levels were found to be consistent with the observed developmental pattern of enzymic maturation in the respiratory and gastrointestinal tract, which play a critical role in the adaptation to the extrauterine environment.

insulin; glucagon; cortisol; triiodothyronine; development; heterochrony

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MAJOR CHANGES in the concentration of enzymes that are specific for the respiratory and gastrointestinal tract are clustered in the perinatal and in the weaning period of the rat (12). Such coordinated developmental changes in the levels of different enzymes point to common control mechanisms. Several studies, including our own (19, 20), have shown that the hormonal regulatory mechanisms that control these changes of the phenotype of the epithelial cells of the gastrointestinal tract are the same in the fetus, neonate, suckling, weanling, and adult.

The rat fetus starts to prepare for birth biochemically and physiologically as soon as organogenesis has been completed. As the interval between the completion of organogenesis and the moment of birth is short, a relatively immature state of the nervous system and skin prevails at birth. Because the biochemical and functional development of the nervous system and skin is a prime parameter for altricial or precocial modes of development, the rat therefore provides a clear example of an altricial mode of development in mammals. In contrast, a precocial mode of development is characterized by a relatively long interval between the completion of organogenesis and the moment of birth. To further elucidate and analyze the role of hormones in the regulation of gene expression in the perinatal period it therefore could be advantageous to complement the altricial "rat model" with a model that exhibits the characteristics of a precocial mode of development. Ideally, this model species should be closely related to the rat and differ in the developmental timing of birth only.

The spiny mouse, *Acomys cahirinus*, appears to meet this requirement, as it also belongs to the taxonomic rodential subfamily of Muridae and, in addition, as it has a long gestation period (39 days) associated with small (2–4) litter size. Organogenesis proceeds somewhat slower in this species than in the rat, requiring 1 additional wk to be completed (25). Neonates show an advanced stage of development, having a fur coat, open eyes, and ears and are capable of locomotion and thermoregulation. Self feeding begins a few days after birth. The rat has developed to a comparable extent in the middle of the 2nd and 3rd postnatal wk, respectively. The late intrauterine developmental stages of the spiny mouse therefore appear comparable to the early extrauterine developmental stages of the rat. In previous studies on the perinatal development of the rat and spiny mouse (22, 23, 29) we have shown that the different developmental timing of birth occurs within the same time course or morphological development as could be monitored by the development of microscopical anatomical features of the organs involved (22, 23, 29). Furthermore, the time course of a number of biochemical parameters of the nervous system could also serve as a reference (9, 24) because it coincides to a large degree with the formation of the morphological architecture. In addition, the completion of organogenesis, allowing enhanced rates of organ-specific enzyme synthesis (33), and the onset of weaning, introducing the adult pattern of organ-specific enzyme levels (19–21), were shown to be developmental landmarks. However, the biochemical and functional development of the respiratory and gastrointestinal tract, both of which derive from the embryonic foregut, shows a well-defined maturation phase in the perinatal period (22–25, 29), the regulation of which appears to be under endocrine control. A comparative study of the circulating levels of glucocorticosteroid and thyroid hormones and of insulin and glucagon in the perinatal period of altricial and precocial species appears to offer a new and unique opportunity to assess the role of these hormones in perinatal maturation, especially because the regulatory mechanisms per se appear to be the same in both model species (22, 23). In this paper we report the developmental profiles of the circulating levels of these hormones in the spiny mouse and compare them with those reported for the rat.

MATERIALS AND METHODS

Animals. Spiny mice (*Acomys cahirinus*) were obtained from our laboratory colony. The animals were fed pelleted rat chow (Hope Farms diet RMHB). Fetal age was read from a chart relating body weight to gestational age (29).

To disturb the animals as little as possible before exsanguination, the cages were located in an easily accessible part of the animal room in the previous evening. Between 9 and 11 A.M. the animals were rapidly anesthesized in their own cages by introducing CO_2 gas and decapitated. This procedure took less than 30 s. Blood was collected from the trunk vessels into heparinized tubes. Aprotonin (Novo) was added to 1,000 $KIU \cdot ml^{-1}$, and the plasma was obtained by centrifugation and stored at −80°C.

Assay of hormone levels. Insulin and glucagon levels were estimated by radioimmunoassay. Pig insulin (Novo) and synthetic glucagon (Serva) were labeled by stoichiometric iodination with chloramine-T (*method A* of Ref. 32). Monoiodinated glucagon and insulin (at the A14 position) were isolated by QAE-Sephadex chromatography (16, 17). Only those fractions that showed the best ratio of binding to a limited and to an excess amount of antibody were used for the assay. The graphs describing the percent tracer bound to the antibody were identical for labeled and unlabeled hormone. The amount of labeled tracer was calculated from the amount of unlabeled hormone necessary to displace 50% of the tracer from the antibody. The specific activity of the labeled hormone was at least 80% of the theoretical specific activity of the homogeneously labeled monoiododerivative of the hormone.

Antibodies against pig insulin were raised in guinea pigs, and blood was obtained by cardiac puncture. In the experiments presented here the C24 serum was used because of the high titer and the volume available. In the radioimmunoassays (RIA) C24 was used in a 1:50,000 (final) dilution. Antibodies against glucagon were raised in both rabbits and guinea pigs, using glucagon covalently linked to thyroglobulin or to bovine serum albumin or in a complex with immunoglobulins as inoculum. Of 23 animals used, only 1 (K63) developed glucagon-specific antibodies, i.e., antibodies that do not cross-react with acid ethanol extract of the ileum (<0.25%). To determine the total amount of glucagonlike-immunoreactivity, serum K12 was used. In the RIAs K63 was used in a 1:2,000 dilution and K12 in a 1:10,000 dilution. The specificity of the three antisera used was further tested immunocytochemically on Bouin-fixed sections (7 μm) of a full-term fetal spiny mouse using the unconjugated immunoperoxidase [peroxidase-antiperoxidase (PAP)] technique (36).

Hormones were assayed in triplicate in 20 μl plasma (insulin) or extract (glucagon), 20 μl antiserum appropriately diluted in RIA buffer, and 160 μl of RIA buffer [0.1 M Na-borate, 0.25% bovine serum albumin (RIA-grade, Sigma), and 60 KIU $aprotonin \cdot ml^{-1}$, final pH = 8.6]. After 24–48 h at 4°C, iodinated insulin (10,000 cpm = 20 pg) or glucagon (5,000 cpm = 5 pg) were added. After another 24 h at 4°C, 80 μl of dialyzed bovine serum and 320 μl of dextran-coated charcoal [1% activated charcoal, 0.25% dextran T-70 in 0.1 M Na-borate (pH = 8.6)] at 4°C were added. After 3 min the suspension was centrifuged and the radioactivity of the immune complexes in the supernatant was estimated by liquid scintillation counting at 56% efficiency. Adding the label 24–48 h after the start of the incubation doubled the sensitivity of the assay (15). The limit of detection for insulin was 10 pg/assay and for glucagon 20 pg/assay. This sensitivity was sufficient to determine circulating insulin levels but not glucagon levels. Therefore, and to remove "big plasma glucagon," the samples for the glucagon RIA were prepared by extracting 500 μl plasma with 1,000 μl acetone for 1 h in the cold in a head-over-head rotator (38). After centrifugation the supernatant was evaporated overnight at 50°C in a vacuum stove and reconstituted in 180 μl RIA buffer. By adding labeled glucagon to 11 different samples from different ages, the recovery of the acetone extraction procedure was calculated to be 82 ± 4%.

Cortisol levels were estimated by high-performance liquid chromatography. Plasma, 100 μl, was applied to a minicolumn containing ~300 mg Extrelut (Merck). After 2 min the column was eluted with 5 ml CH_2Cl_2 that had been purified by Al_2O_3 absorption. The CH_2Cl_2 was evaporated in a stream of N_2 gas at 40°C. The recovery of cortisol in this procedure was more than 90% (34). The residue was dissolved in 100 μl isooctane:chloroform: methanol:water (25:71:3.75:0.25) and injected into a glass capillary column (150 × 3.4 mm) filled with Si 60 (5 μm, Merck). The column was eluted at 1 ml/min with the same solvent mixture (3). The effluent was monitored at 254 nm. Pure cortisol had a typical retention time of 5.5 min. The limit of detection was 2.5 ng cortisol. Corticosterone levels were estimated by the same HPLC procedure except that the solvent mixture was isooctane:chloroform:methanol:water (56.5:40.5:2.9:0.12) (3).

Thyroxine (T_4) and triiodothyronine (T_3) levels were estimated by RIA. T_4 and T_3 were labeled with ^{125}I by the chloramine-T method and purified by elution with 10 mM NaOH from a Sephadex G-25 (Pharmacia) column (18). Antibodies against T_4 were obtained by vaccinating rabbits with bovine thyroglobulin (T_4 antiserum

3045) or with T_3 bovine thyroglobulin conjugate (T_3 antiserum 137) and used in a final dilution of 1:8,000 and 1:140,000, respectively. At 50% binding of the tracer the anti-T_4 antibody cross-reacted 0.2% with T_3, while the anti-T_3 antibody cross-reacted <0.1% with T_4.

Hormones were assayed in 20 μl plasma, 100 μl antiserum, appropriately diluted in RIA buffer, 50 μl tracer, containing 10,000 cpm (~2 pg hormone), and 830 μl of RIA buffer [75 mM Na-barbital, 0.25% bovine serum albumin (pH = 8.6)]. Standard curves were prepared in iodothyronine-free plasma, that was prepared by repeated washings of a homologous plasma pool with AG1 X2 resin (Bio-Rad) equilibrated with Na-barbital (pH = 8.6). To displace thyroid hormones from endogenous thyroid hormone-binding proteins, 300 μg anazolene sodium (T_4) or 3 mg Na salicylate (T_3) were added to the reaction vessels. After 24 h of incubation at 4°C an equal volume of 30% polyethylene glycol 6,000/4% bovine serum was added. After an additional 15 min at 4°C the suspension was centrifuged and radioactivity in the immune complexes was estimated by γ-scintillation counting. The limit of detection was 20 pg T_4 and 2 pg T_3 per assay.

The graphs depict means ± SE of hormone assays of three to four different plasma samples from as many different litters. In case only two samples were available, the average is shown. Most estimations of glucagon and insulin levels were derived from the same plasma samples. Only these were used for calculation of the insulin-to-glucagon ratio. The glucagonlike immunoreactivity-to-glucagon (gli/glu) ratio was calculated only for those plasma samples in which both glu and gli were measured. Due to the size of the animals, thyroid hormones and cortisol had to be estimated in separate plasma samples.

Statistical analysis. All statistical comparisons have been based on polynomial regression analysis and analysis of variance techniques, using the BMDP statistical packages from the Health Sciences Computing Facility of the University of California, Los Angeles (1983). For these comparisons, the experimental data were logarithmically transformed.

RESULTS

Glucagon and insulin. The levels of these peptide hormones were assayed by RIA. The ^{125}I-labeled hormones were pig insulin and glucagon. To validate the RIA procedure, several controls were made. In the first place, the changes in insulin and glucagon levels in the perinatal period of the rat were assayed and were found to be completely comparable with those reported in the literature (not shown). Furthermore, it was ascertained that the (specific) antibodies reacted only with acid ethanol extracts of the pancreas and not with such extracts from other parts of the gut of the spiny mouse. In addition, parallel displacement curves were obtained when the antibodies were incubated in the presence of tracer and increasing amounts of pancreatic extracts and pig insulin or glucagon, respectively (not shown). As a final control the antibodies were tested for immunocytochemical staining of the islets in the pancreas and of the gut of the spiny mouse (Fig. 1). Figure 1, *A* and *C*, shows that, as in the rat, the glucagon-producing cells are found in the periphery of the islets, while Fig. 1*E* shows that insulin-producing cells are found in the center of the islets. Furthermore, Fig. 1, *B* and *D*, shows that serum K63 reacts with pancreatic glucagon only, while serum K12 does not discriminate between glucagon and enteroglucagon.

Figure 2 shows the developmental profile of plasma glucagon levels of the spiny mouse, expressed as picomole equivalents per liter. Before birth average levels fluctuate around 60 pM. At birth glucagon levels almost double to ~100 pM and remain high for the next 10 days. Thereafter levels decrease to ~70 pM between 2.5 and 4 wk after birth. In adults, average levels amount to ~60 pM (200 pg/ml). Multiple comparisons analysis of variance shows that the differences in glucagon concentration at different ages are statistically significant ($p = 0.04$). Figure 2 also shows the change in circulating levels of gli of acetone extracts of plasma. After birth, the level of (extrapancreatic) gli is low as exemplified by a ratio of gli/glu that approximates 1. Before birth, the levels of circulating gli are elevated: at 9 days before birth there is 3 times as much gli than glucagon. Multiple comparisons analysis of variance shows that the differences in gli/glu at different ages are statistically significant ($P = 0.05$).

Figure 3 shows the developmental profile of plasma insulin levels of the spiny mouse, also expressed as picomole equivalents per liter. Up to 5 days before birth insulin levels fluctuate between 110 and 160 pM. Then levels almost double to ~220–250 pM between 3 days before and 1 day after birth. Thereafter average insulin levels steeply decrease to between 100 and 150 pM. In adults, average levels amount to almost 200 pM (1.2 ng/ml). The developmental profile of insulin fits a 7th degree polynomial ($P = 0.01$).

Cortisol. Preliminary determinations using high-pressure liquid chromatography (HPLC) analysis indicated that cortisol rather than corticosterone was the circulating glucocorticosteroid in spiny mice. The peak was shown to be absent in the plasma of three males that were adrenalectomized and gonadectomized 7 days previously. Moreover, it was shown in three additional males to increase to a value of 3,150 ± 220 nM 30 min after an intraperitoneal injection of 125 μg ACTH-(1—24) (Cortrosyn, Organon). To further prove that this peak was identical with that of cortisol, the HPLC-purified cortisol-containing fractions of one of the adrenalectomized spiny mice, of one of the ACTH-(1—24)-injected spiny mice, and of an untreated 13-day-old animal were subjected to mass spectroscopy. By comparison with a standard cortisol preparation the presence and identity of cortisol in these plasma fractions was confirmed by the typical ratio's of M/Z peaks at 515, 605, and 636 (not shown).

Figure 4 shows the developmental profile of plasma cortisol levels in the spiny mouse. Seven to eight days before birth, cortisol levels are very high (~350 nM). Subsequently, levels decrease to ~100 nM at birth and remain at that level for the next 3–4 days. In the next 10

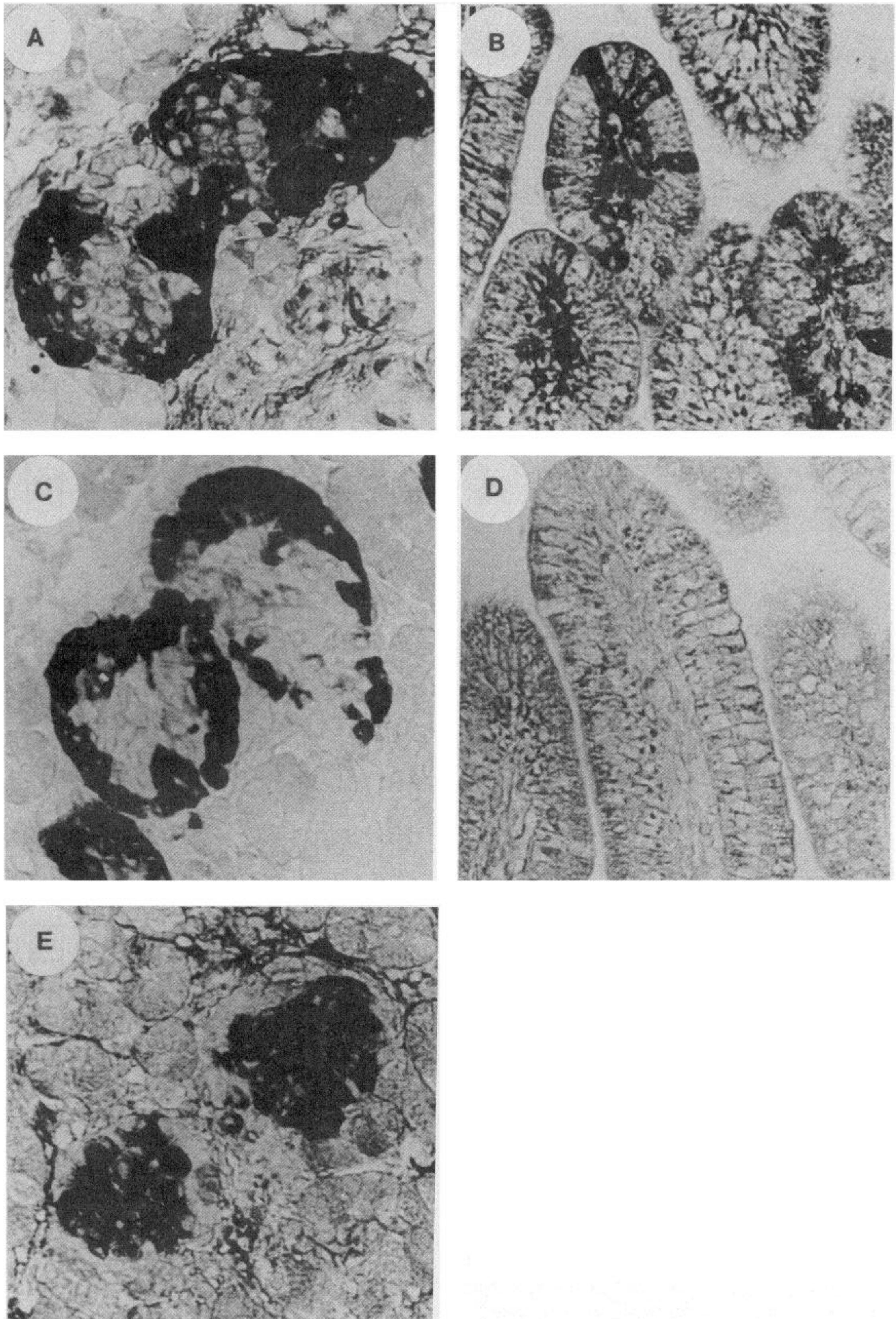

FIG. 1. Immunocytochemical staining of pancreas (*A*, *C*, and *E*) and gut (*B* and *D*) of a full-term spiny mouse fetus (×400). In Fig. 1, *A* and *B*, sections (7 μm) were incubated with serum K12, that does not discriminate between pancreatic and gut glucagonlike immunoreactivity. In Fig. 1, *C* and *D*, sections were incubated with serum K63, that is, specific for pancreatic glucagon; and in Fig. 1*E* section was incubated with serum C24 that is specific for pancreatic insulin.

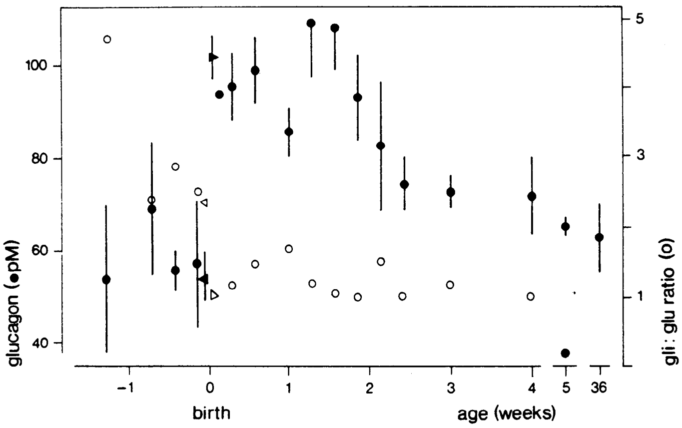

FIG. 2. Developmental changes in level of glucagon in blood plasma of spiny mouse (*closed symbols*). On *horizontal axis*, age of animals is indicated; on *vertical axis*, glucagon concentration is depicted in pM equivalents of pig glucagon. No distinction of sex is made. On *day 0, triangles* pointing to *left* represent fetuses on day of expected delivery, while *triangles* pointing to *right* represent spontaneously born neonates. Furthermore, ratio of glucagonlike immunoreactivity (gli; measured with serum K12) and glucagon (glu; measured with serum K63) is depicted as *open circles*.

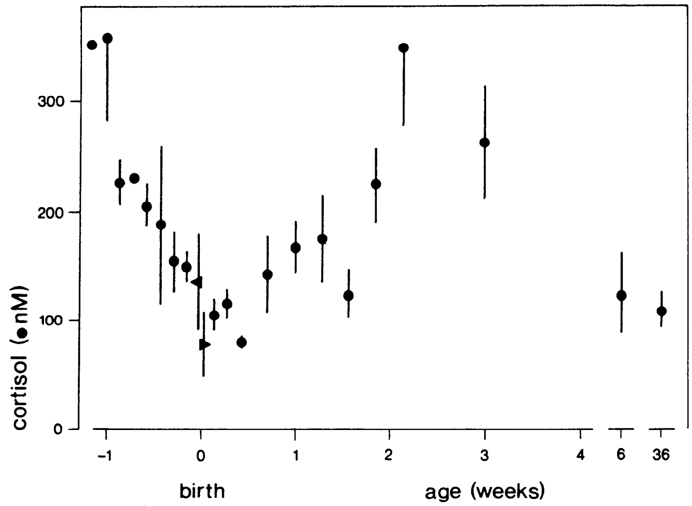

FIG. 4. Developmental changes in levels of cortisol in blood plasma of spiny mouse. On *horizontal axis*, age of animals is indicated; on *vertical axis* cortisol concentration is indicated. No distinction of sex is made. On *day 0, triangles* pointing to *left* represent fetuses on day of expected delivery, while *triangles* pointing to *right* represent spontaneously born neonates.

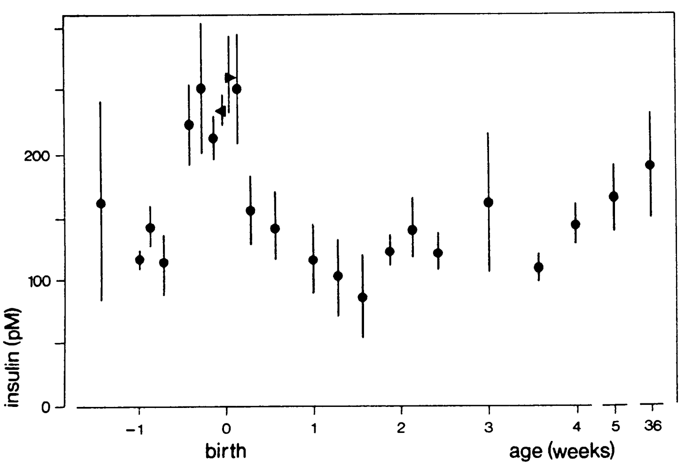

FIG. 3. Developmental changes in level of insulin in blood plasma of spiny mouse. On *horizontal axis*, age of animals is indicated; on *vertical axis* insulin concentration is depicted as pM equivalents of pig insulin. No distinction of sex is made. On *day 0, triangles* pointing to *left* represent fetuses on day of expected delivery, while *triangles* pointing to *right* represent spontaneously born neonates.

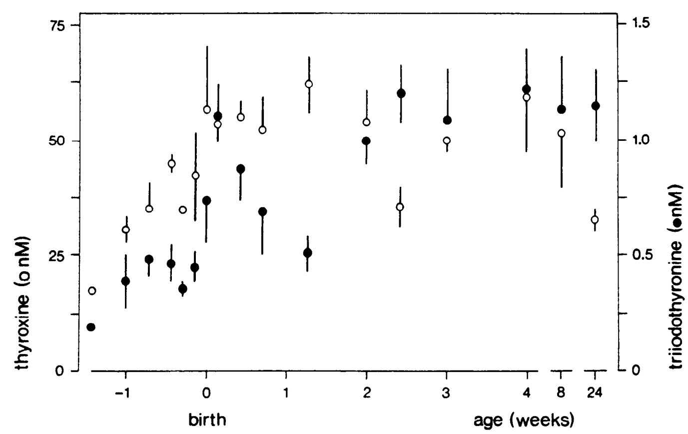

FIG. 5. Developmental changes in levels of thyroxine (*open symbols*) and triiodothyronine (*closed symbols*) in blood plasma of spiny mouse. On *horizontal axis*, age of animals is indicated; on *vertical axis* thyroid hormone concentration is indicated. No distinction of sex is made.

days cortisol levels increase to over 300 nM in 15-day-old animals. Thereafter levels decrease. In adults, average levels amount to ~100 nM (40 ng/ml). The developmental profile of cortisol fits a 4th degree polynomial ($P = 0.007$). Corticosterone was neither detectable (<5 nM) when cortisol levels were high nor when they were low.

Thyroxine and triiodothyronine. Figure 5 shows the developmental profile of T_4 and T_3 in the plasma of the spiny mouse. In the fetus, T_4 levels increased from 15 nM at 10 days before birth to ~60 nM at birth. During the first 8 postnatal wk, T_4 concentrations remained at the same level. In 6-mo-old animals, the level decreased to ~35 nM (30 ng/ml). During the last week before birth, T_3 levels were ~0.4 nM. Between 1 day before and 1 day after birth, levels increased 2.5-fold to 1.1 nM. Average T_3 levels transiently decreased to 0.5 nM at 9 days after birth. Then levels increased again to ~1.2 nM at 17 days and remained constant thereafter (0.8 ng/ml). Multiple comparisons analysis of variance shows that the differences in T_3 concentration at different ages are statistically significant ($P = 0.0001$).

DISCUSSION

Developmental profiles in rat and spiny mouse. The simultaneous presence within one taxonomic group of altricial and precocial patterns of development, i.e., with a different development timing of birth, is extremely rare (7). Such a different developmental timing of the same process is called heterochrony (11) and offers a good opportunity for a comparative study of the role of regulatory factors, that are involved in the coordination of perinatal adaptation. Schematic graphs of the levels of glucocorticosteroid and thyroid hormones and of the insulin-to-glucagon ratio as a function of development of rat and spiny mouse are shown in Figs. 6–8. The ages for both species have been aligned in such a way that the developmental stages match (29).

Figure 6 shows a comparison between developmental changes in the corticosterone levels in the rat (derived from Refs. 14 and 27) and cortisol levels in the spiny mouse. Although total instead of free glucocorticosteroid levels are compared in Fig. 6, changes in free levels (14) and in organ influx (30) often reflect changes in total levels. The curves (Fig. 6) suggest that glucocorticosteroid synthesis starts before birth at approximately the same developmental stage, i.e., 5–6 days before birth in the rat and approximately 2 wk before birth in the spiny mouse. Although total glucocorticosteroids in the rat attain peak levels at 3 days before birth and then rapidly decrease to low levels at birth, a similar but more protracted profile is seen in the spiny mouse. A prenatal peak in glucocorticosteroid levels appears to be a prerequisite for perinatal adaptation in mammalian species (26). Both rat and spiny mouse show a second maximum around weaning.

Because T_3 is metabolically more active than T_4, Fig. 7 shows a comparison between the developmental changes in total T_3 levels in rat (derived from Refs. 6 and 13) and in spiny mouse. Although total T_3 concentrations were used for the comparison as only these were measured in the spiny mouse, changes in free T_3 levels (39) and in organ influx (31) reflect changes in total T_3 levels in the rat. Although rat and spiny mouse show virtually identical increases in T_3 levels around weaning, the substantial increase in T_3 concentration at birth is only present in the spiny mouse. This postpartum T_3 increase so far has only been reported in species with relatively long fetal periods such as ruminants (8, 37) and primates (1, 18). Although an important role for T_3 has been implicated in perinatal adaptation (8, 37), its

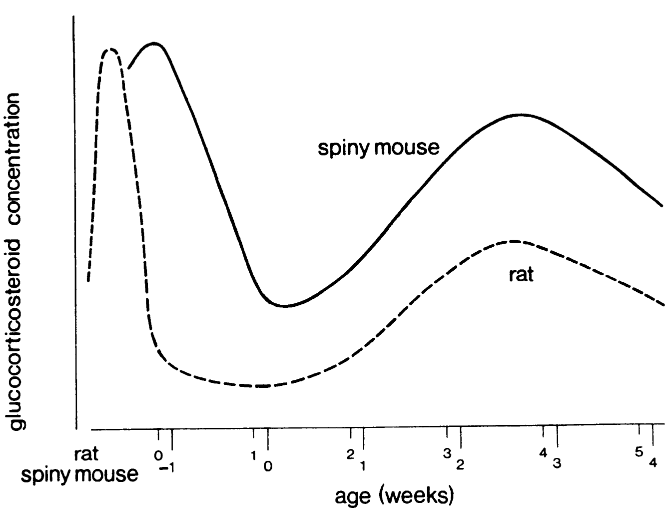

FIG. 6. Schematic developmental profiles of rat corticosterone (- - -) and spiny mouse cortisol (——) levels. On *horizontal axis*, ages for both species have been aligned in such a way that developmental stages match (29). Birth is indicated by *0*. Note that in both species onset of increases in glucocorticosteroid levels occurs at same developmental time point, that immediate prenatal period is characterized by decreasing hormone levels, and that immediate postnatal period is characterized by a low level of glucocorticosteroid hormone. This period lasts appreciably longer in rat than in spiny mouse.

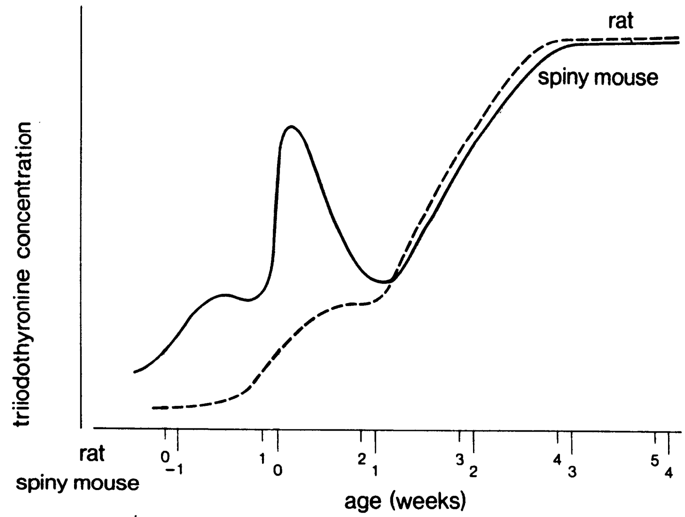

FIG. 7. Schematic developmental profiles of rat (- - -) and spiny mouse (——) triiodothyronine levels. On *horizontal axis*, ages for both species have been aligned in such a way that developmental match (29). Birth is indicated by *0*. Note that a substantial increase in triiodothyronine levels at birth occurs in spiny mouse only; and that onset of weaning increase in hormone levels occurs at same developmental time point in both species.

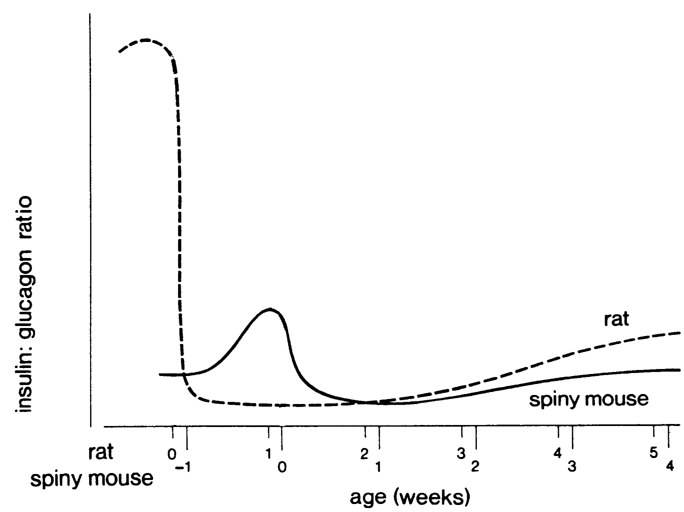

FIG. 8. Schematic developmental profiles of rat (- - -) and spiny mouse (——) molar insulin; glucagon ratio. On *horizontal axis*, ages for both species have been aligned in such a way that developmental stages match (23). Birth is indicated by *0*. Note that an acute decrease in insulin glucagon ratio occurs around birth only and that an prenatal peak in insulin glucagon ratio is much smaller in spiny mouse than in rat.

significance remains to be established in view of its virtual absence in altricial species and its presence in closely related species as shown here.

A comparison of changes in the ratio of glucagon and insulin levels in the perinatal period of rat and spiny mouse is shown in Fig. 8. We have depicted this ratio because it appears to govern intracellular cAMP levels in the liver in vivo (5, 35) that in turn correlate well with the observed adaptational changes in enzyme levels (4). The predominant changes in the insulin-to-glucagon ratio of the rat (derived from Refs. 2 and 10 and Lamers and Mooren, unpublished observation) and of the spiny mouse occur around birth, being high in the full-term fetus and decreasing abruptly after birth. Although in the spiny mouse the insulin and glucagon levels are based on pig insulin and glucagon as standards, the maximum levels and ratio are probably approximately threefold lower than in the rat and are in the range observed for the human fetus and neonate (10). The very high insulin-to-glucagon ratio in the rat may be due to complexed insulin in the rat fetus (40). In both species, the insulin-to-glucagon ratio increases at weaning, although the levels that are attained are lower than those observed in the full-term fetus.

Perinatal regulation of enzyme levels. From these and previous comparisons (22–25, 29) between rat and spiny mouse a few common features can be discerned that appear relevant for the regulation of enzyme levels in the perinatal period. The prenatal period is characterized by high levels of glucocorticosteroid and a high insulin-to-glucagon ratio, leading to an increase in many organ-specific enzymes in the gastrointestinal tract to only 20–50% of maximal adult values. On the other hand the preweaning period is characterized by high levels of glucocorticosteroids and a low insulin-to-glucagon ratio, leading to much higher or even maximal adult values. Finally, enzymes that show increases in activity in the prenatal and preweaning period, show a much more extensive increase in the neonatal period in the rat than in the spiny mouse; this may be related to the much more dramatic drop in the insulin-to-glucagon ratio at birth. The importance of developmental changes in T_3 levels for developmental changes in enzyme levels is not easily assessed, as T_3 levels increase at birth in the spiny mouse but not in the rat without apparently affecting perinatal adaptation in either species. Furthermore, enzymes that, like T_3, increase in level during the preweaning period are only marginally affected by thyroid hormone (20, 22, 23). It is therefore not surprising that the regulatory role of thyroid hormone in gene expression is still poorly understood (28).

The human condition: altricial and precocial. On balance, Figs. 6–8 show that the spiny mouse condition reflects the human condition in the perinatal period more adequately than the rat condition. In other words, the perinatal changes in hormone concentrations and the perinatal changes in organ-specific enzyme levels in the respiratory and alimentary tract (22, 23, 29) indicate that, as far as these properties are concerned, the human being has a precocial mode of development. This precocial mode of development, in combination with the altricial mode of development of the central nervous system, makes primates rather unique and illustrates that most model systems when studied alone are inadequate. However, studies in the rat-spiny mouse model system promise to be very useful for the analysis of regulatory factors in perinatal adaptation as they allow the discrimination of environmental (intrauterine) factors from developmental (fetal) factors (23). This inexpensive, heterochronic (11) model system therefore allows more adequate and general predictions on fetus and environment than either system alone and so far has been able to explain perinatal adaptational processes in the human fairly adequately.

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